THE IMAGE OF A UNIVERSALLY CLOSED STACK UNDER A SURJECTIVE MORPHISM IS UNIVERSALLY CLOSED, AND PROPER WHEN IT IS SEPARATED OF FINITE TYPE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-algebraic-stacks-morphisms/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The definition of a morphism of algebraic stacks being universally closed through the closedness of the induced maps on topological spaces after base change. The author's section on universally closed morphisms, in which the counted statement occurs. Formulates what has to be checked for the target of the surjection.
- The stability of surjectivity under base change. Invoked inside the counted proof at source lines 8712-8715 as a cross-chapter pointer. Makes the base changed map surjective so that preimages of closed sets can be transported.

Source reference. Original reference stacks-properties-lemma-base-change-surjective: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2031–2031. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-base-change-surjective, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{Z}$ be an algebraic stack. Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks over $\mathcal{Z}$. If $\mathcal{X}$ is universally closed over $\mathcal{Z}$ and f is surjective then $\mathcal{Y}$ is universally closed over $\mathcal{Z}$. In particular, if also $\mathcal{Y}$ is separated and of finite type over $\mathcal{Z}$, then $\mathcal{Y}$ is proper over $\mathcal{Z}$.*

Proof. Assume $\mathcal{X}$ is universally closed and f surjective. Denote $p : \mathcal{X} \to \mathcal{Z}$, $q : \mathcal{Y} \to \mathcal{Z}$ the structure morphisms. Let $\mathcal{Z}' \to \mathcal{Z}$ be a morphism of algebraic stacks. The base change $f' : \mathcal{X}' \to \mathcal{Y}'$ of f by $\mathcal{Z}' \to \mathcal{Z}$ is surjective (Properties of Stacks,

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Lemma stacks-properties-lemma-base-change-surjective) and the base change $p' : \mathcal{X}' \to \mathcal{Z}'$ of p is closed. If $T \subset |\mathcal{Y}'|$ is closed, then $(f')^{-1}(T) \subset |\mathcal{X}'|$ is closed, hence $p'((f')^{-1}(T)) = q'(T)$ is closed. So q' is closed. $\square$